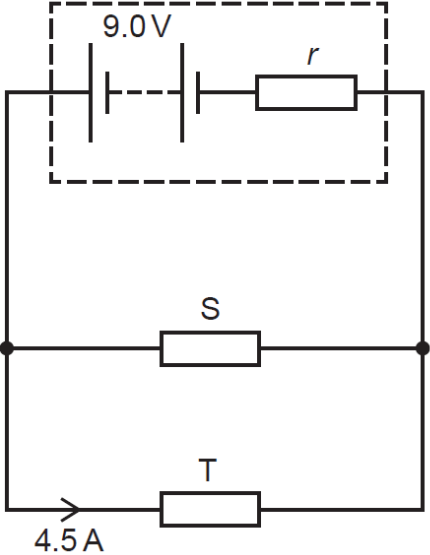


4	(a)	A battery of electromotive force (e.m.f.) 9.0 V and internal resistance r is connected to two resistors S and T, as shown in Fig. 4.1.  Fig 4.1 A total charge of 650 C moves through resistor S in a time interval of 540 s. During this time resistor S dissipates 4800 J of energy. The current in resistor T is 4.5 A. Assume that the e.m.f. of the battery remains constant. Calculate:
	(i)	the current in resistor S
		current =A [1]
		Solution: $I = \frac{Q}{t} = \frac{650}{540} = 1.2 \text{ A}$ 1
	(ii)	the internal resistance r of the battery.
		$r = \dots\dots\dots \Omega$ [3]
		Solution: $V = \frac{W}{Q} = \frac{4800}{650} = 7.38 \text{ V or use } V = \frac{P}{I} \text{ and } P = \frac{W}{t}$ 1

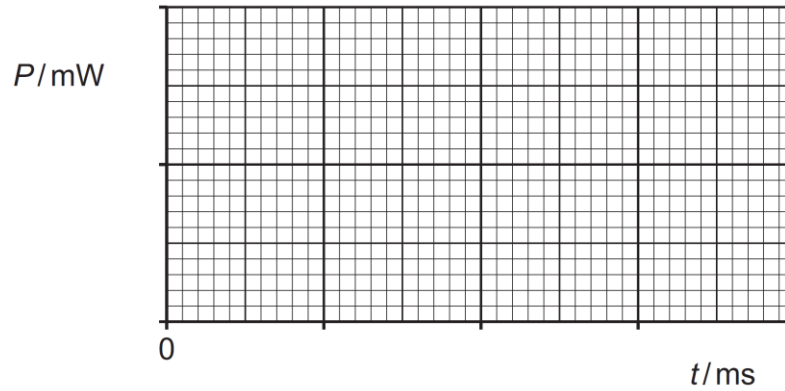
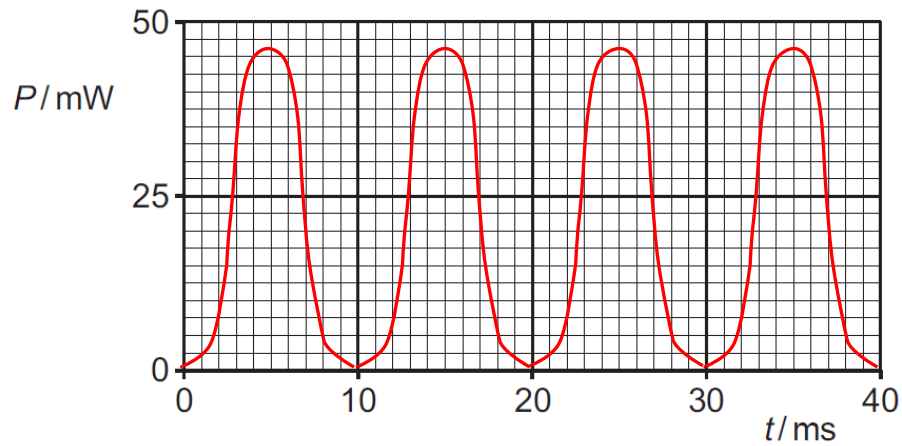


Fig. 4.2

[3]

Solution:1
1
1